



INTRODUCTION TO BUILDING MATERIALS - BRICKS

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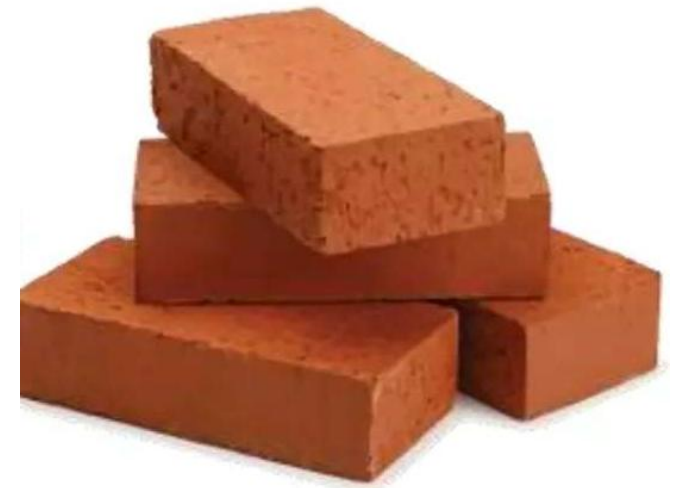
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Bricks - Definition

- ❖ A brick is a small rectangular building unit made from clay, shale, concrete, fly ash, or similar materials, and is used in the construction of walls, foundations, pavements, and other structures
- ❖ Standard brick dimensions include:
 1. Traditional Brick: 23 cm (length) $\times$ 11.4 cm (width) $\times$ 7.6 cm (height)
 2. Modular Brick: 19 cm (length) $\times$ 9 cm (width) $\times$ 9 cm (height)





Composition of Bricks

A. Silica (SiO_2) – 50% - 60%

- ❖ Free silica (sand) is a main constituent, if added to clay in suitable proportion makes hard and prevents it from warping and shrinkage on drying.
- ❖ Silica, if present in greater proportion, makes a brick brittle. Silica present in the combined form (aluminum silicate) does not form good bricks, as it will shrink and develop cracks.



Composition of Bricks

B. Alumina (Al_2O_3) - 20% - 30%

- ❖ Alumina is one of a main constituent of every clay. Loam soil (adhesive soil) forms good clay. In absence of sand, pure clay will develop cracks due to shrinkage on drying and burning. A good clay bricks should contain about 20% of alumina.
- ❖ Alumina absorbs water and renders the clay plastic. If alumina is present in excess of the specified quantity, it produces cracks in brick on drying. Clays having exceedingly high alumina content are likely to be very refractory. Both silica and alumina should be in free form.



Composition of Bricks

C-Lime (CaCO_3) – 5% - 6%

- ❖ Normally constitutes less than 10 percent of clay. This also should be present in small quantities in the brick earth. It should be in a finely produced condition and it should not be in the form lumps or clods. Lime prevents shrinkage of raw bricks on drying.
- ❖ It helps fusion of sand at the kiln temperature. This fused sand will bind the bricks particles fast.



Composition of Bricks

D- Iron oxide:

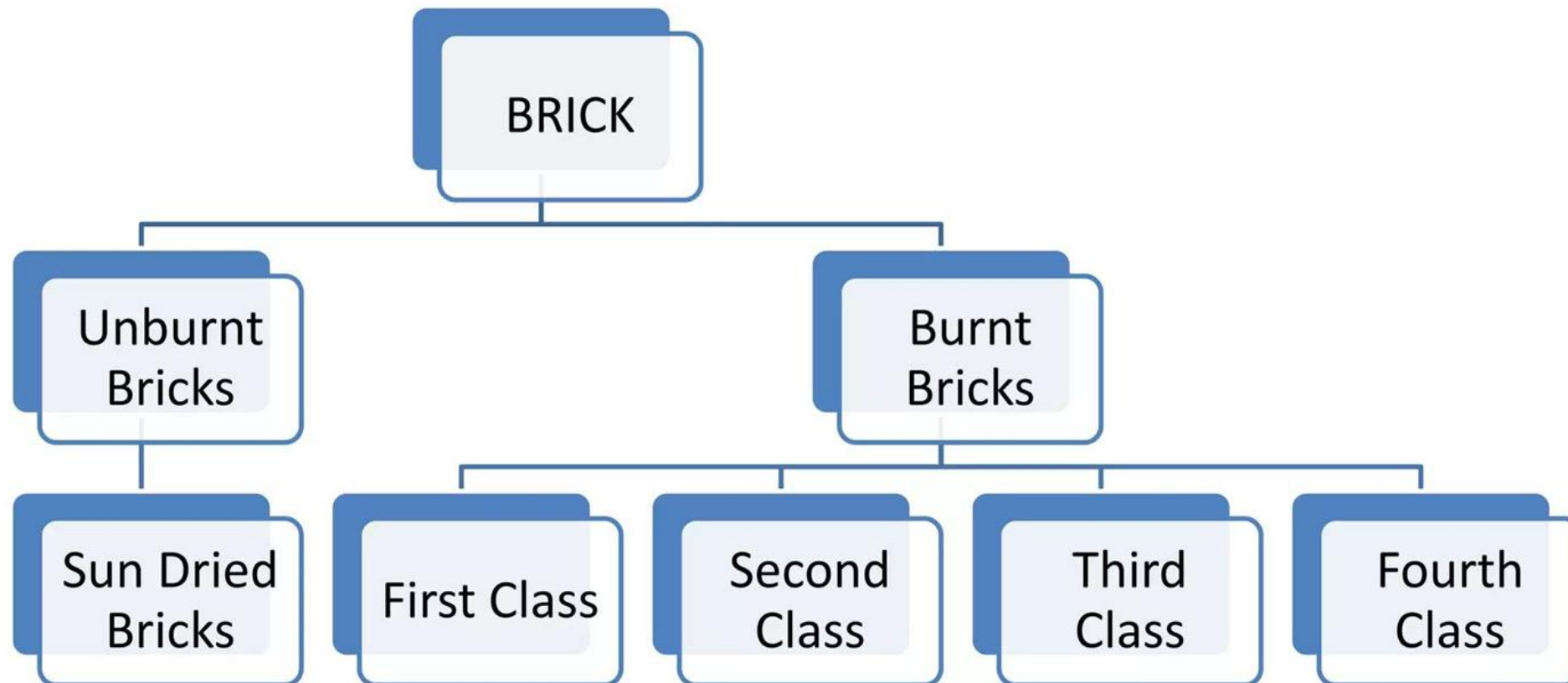
A small quantity of oxide of iron (5-6%) is desirable. It helps the fusion of sand like lime. It gives red colour to burn bricks. Excess of iron oxide imparts dark blue or blackish colour to brick, while, a lower percentage of iron oxide makes the brick yellow in colour. Iron oxide makes the bricks hard and strong.

E. Magnesia:

Rarely exceeding 1 percent affects the colour and makes the brick yellow. A small amount of magnesia helps to decrease the shrinkage of bricks. This gives a yellow tint to the bricks. But excess of magnesia is not desirable as it tends to produce the decay of bricks.



Classification of Bricks





First Class Bricks

- ❖ These Bricks are well burnt, having smooth & even surface, with perfect rectangular shape & uniform reddish colour. When you stuck each other, these give a metallic ringing sound.
- ❖ These should not absorb water more than 20% of its weight when immersed in cold water for 24 hrs.
- ❖ Crushing Strength less than in 55 kg/cm^2 is not allowed. Minimum Compressive strength is 35 kg/cm^2 .
- ❖ They are used as good structure as outer walls and facing works when no plastering is done. These are also used in floors and reinforced bricks slab. Such bricks should be laid in rich mortar.





Second Class Bricks

- ❖ They are not perfectly rectangular in shape and are having rough surface, but are hard, slightly over burnt and uniform in colour.
- ❖ These give ringing sound when struck with each other.
- ❖ Water absorption is more than 22% by weight, when immersed in water for 24 hrs.
- ❖ These are used for internal walls, not exposed to atmosphere, these are used in facing work, which should be plastered.
- ❖ These can not be used for R. B. Work. Such bricks may be laid in mud or lime mortar.





Third Class Bricks

- ❖ These bricks are also known as “Pila Bricks”. These are not burnt properly in the kilns & may be slightly under burnt, hence these are soft & can be easily broken.
- ❖ These are light red in colour, with yellowish tinge.
- ❖ On striking, these do not give a ringing sound.
- ❖ These should not absorb more than 25% of water by weight when placed in cold water for 24 hrs.
- ❖ These are used for inferior construction works, or at place when there is less rain fall or presence of dampness.





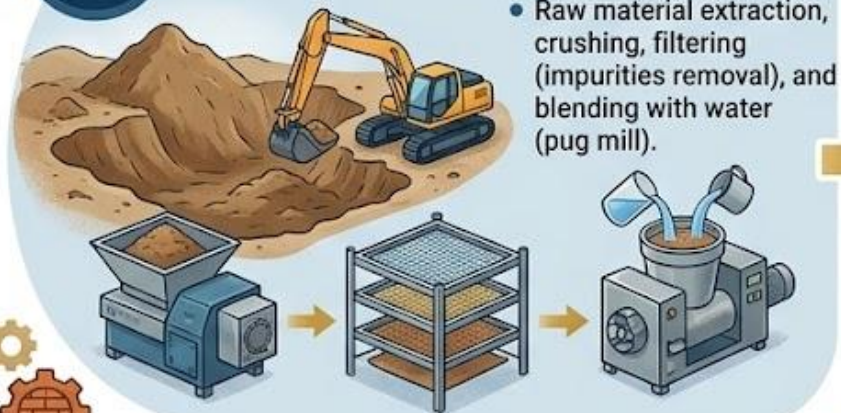
Fourth Class Bricks

- ❖ It is also known as “Jhama and Over Burnt Bricks”.
- ❖ Due to excess fusion & temperature bricks are over burnt, loose their shape & get twisted.
- ❖ These bricks are dark bluish in colour.
- ❖ These bricks are not used in brick work. They are used in the form of broken pieces, these may be used as road metal, also in foundation and floors as soling material.

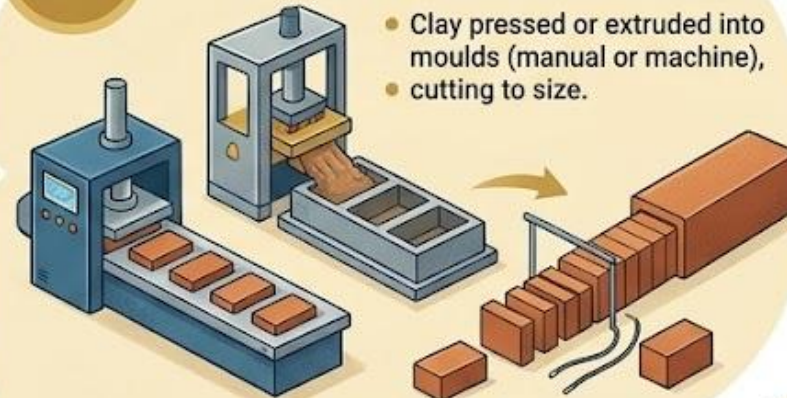


THE COMPLETE BRICK MANUFACTURING PROCESS

1. PREPARATION OF CLAY



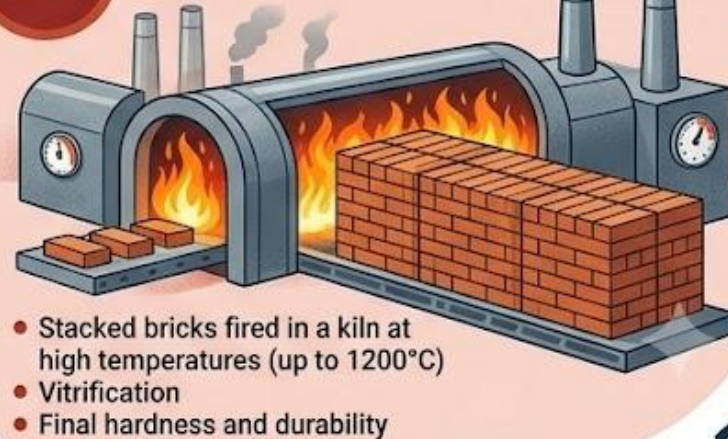
2. MOULDING



3. DRYING



4. BURNING





Brick Manufacturing

Ground Molded Bricks



Fig.: Ground moulding

Table Moulded Bricks



Fig.: Table moulding



Brick Manufacturing

Machine Moulding



Fig.: Machine moulding



Brick Manufacturing

Drying of bricks



Fig.: Drying of Bricks



Brick Manufacturing

Burning

Clamps



Kilns



PREPARATION OF CLAY FOR BRICKS: — A STEP-BY-STEP PROCESS —

1. MINING & EXCAVATION



Extract raw clay from deposits; Remove overburden.

2. WEATHERING



Expose clay to elements; Break down large clumps naturally.

3. CRUSHING & GRINDING



Pulverize the weathered clay into fine particles.

4. SCREENING & CLEANING



Remove impurities like pebbles, roots, and organic matter.

5. TEMPERING & BLENDING



Add water and other materials to achieve desired plasticity and composition.

6. PUGGING & KNEADING



Thoroughly mix and knead to remove air pockets; Form workable plastic mass.



QUALITY OF GOOD BRICKS

- ❖ The bricks should be table moulded, well burnt in kilns, free from cracks and with square edges. The colour should be uniform and bright.
- ❖ The brick should give a clear ringing sound when struck with each other.
- ❖ The brick when broken or fractured should not show a bright homogeneous and uniform compact structure free from voids.
- ❖ The brick should be sufficiently hard. No impression should be left on the brick surface when it is scratched with a fingernail.
- ❖ The brick should not break into pieces when dropped flat on hard ground from a height of about one meter.
- ❖ The brick, when soaked in water for 24 hours, should not show deposits of white salts when allowed to dry in shade.
- ❖ The brick should not absorb more than **20%** water by weight for **first-class bricks** and **22%** by weight for **second-class bricks** when soaked in cold water for 24 hours.
- ❖ No brick should have a crushing strength below **5.50 N/mm²**.



TEST ON BRICKS – Water Absorption

- ❖ Five bricks are taken for the test. They are allowed to dry in an oven at **110°C to 115°C** until they attain a constant weight, which usually takes **48 hours**.
- ❖ They are then cooled at room temperature, which generally takes **4 to 6 hours** without using a fan, and the dry weight **W₁** is measured.
- ❖ The bricks are then kept in clean water at **29°C** for **24 hours**, wiped dry with a damp cloth, and the wet weight **W₂** is measured.
- ❖ The average percentage of water absorbed, expressed as a percentage of the dry weight, is reported.

$$\text{Water Absorption (\%)} = \frac{W_2 - W_1}{W_1} \times 100$$

Where:

W₁ = Dry weight of the brick after oven drying

W₂ = Weight of the brick after 24 hours of water immersion



TEST ON BRICKS – Compressive Strength

- ❖ **Step 1** – A brick is taken and the two parallel sides are made smooth. Then, immerse the brick in water for 24 hours. Take out the bricks from water. Wipe out the surface of the brick using a clean cloth.
- ❖ **Step 2** – Fill the frog with cement mortar. Now, place the bricks under damp jute bags for 24 hours. Then, place the bricks in water at room temp for about 3 days.
- ❖ **Step 3** – Take out the bricks from water. Wipe the brick surface with a cloth.
- ❖ **Step 4** – Place the specimen in Compression Testing Machine in such a way that the frog is at the top. Place two numbers of 3 mm thick plywood above and below the brick to apply uniform load on the specimen.
- ❖ **Step 5** – Apply the loading at a constant rate of 14 N/mm^2 per minute until failure.
- ❖ **Step 6** – The maximum load at failure divided by the area of the bed face gives the compressive strength of brick.
- ❖ If the compressive strength of any specimen falls below 20% of the minimum average compressive strength specified for that class of brick, then that particular test result is not considered.
- ❖ Minimum Compressive Strength of brick required to be used for construction is 3.5 N/mm^2 .



TEST ON BRICKS – Efflorescence

- ❖ **Step 1** – Place the bricks in a porcelain or glass dish of 200mm diameter and 40 mm deep. Fill the dish with distilled water to a depth of 25 mm. Maintain the distilled water at room temperature ($27^{\circ}\text{C} \pm 2^{\circ}\text{C}$).
- ❖ **Step 2** – Place the brick in water until water gets absorbed by the brick or gets evaporated completely. Fill the water again up to a depth of 25mm and wait until the water disappears completely.
- ❖ **Step 3** – Observe the deposit of efflorescence on the surface of the brick. The efflorescence is classified as:
 - ❖ *Nil* – If the deposit of efflorescence is imperceptible.
 - ❖ *Slight* – If the deposit of efflorescence covers less than 10% of exposed area of brick.
 - ❖ *Moderate* – If the deposit of efflorescence covers 10% to 50% of exposed area of brick.
 - ❖ *Heavy* – If the deposit of efflorescence covers more than 50% of exposed area of brick but the deposits do not flake away from brick surface.
 - ❖ *Serious* – If the deposit of efflorescence covers more than 50% of exposed area of brick and the deposits flake away from brick surface.



TEST ON BRICKS – Shape & Size Test

- ❖ The brick should be of **standard size**, and its shape should be **truly rectangular with sharp edges**.
- ❖ **Twenty (20) bricks** of standard size (**190 mm × 90 mm × 90 mm**) are selected at random.
- ❖ The bricks are stacked:
 - ❖ **Lengthwise**
 - ❖ **Widthwise**
 - ❖ **Heightwise**
- ❖ The overall dimensions of the stack should be within the following limits:

Dimension	Acceptable Range
Length (20 bricks)	3680 mm – 3920 mm
Width (20 bricks)	1740 mm – 1860 mm
Height (20 bricks)	1740 mm – 1860 mm





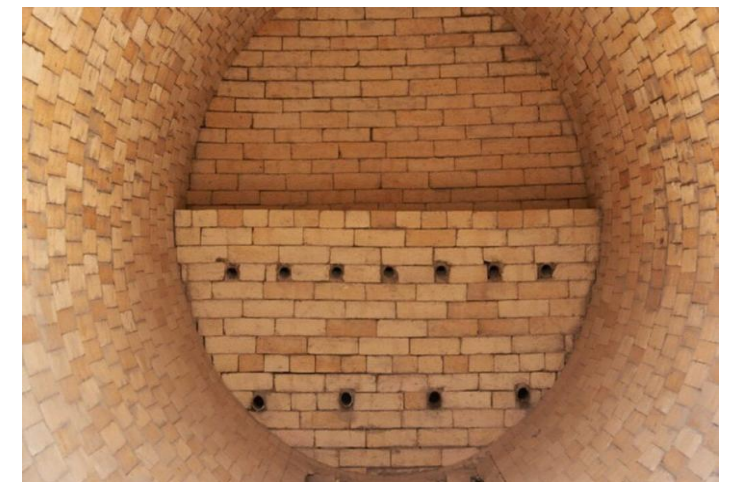
Classification of Special Use Bricks

- ❖ **Refractory Bricks** – High heat resistance
- ❖ **Concrete Blocks** – Structural and modular construction
- ❖ **Paver Blocks** – Decorative and load-bearing surfaces
- ❖ **AAC Blocks** – Lightweight and thermally insulated
- ❖ **CLC Blocks** – Lightweight and eco-friendly



Refractory Bricks – Introduction

- ❖ Refractory bricks are heat-resistant building materials designed to withstand very high temperatures (1000°C – 1800°C or more) without losing their strength or shape.
- ❖ They are manufactured from refractory materials such as fire clay, silica, magnesia, alumina, and chromite.
- ❖ These bricks have low thermal conductivity, high mechanical strength, and excellent resistance to thermal shock and chemical attack





Concrete Blocks - Introduction



Made of cement, sand, aggregates, and water.



Used in load-bearing and partition walls.



Larger in size than traditional bricks – faster construction.





Concrete Blocks - Types & Benefits

❖ Types:

- Solid concrete blocks
- Hollow concrete blocks

❖ Benefits:

- Economical
- Fire-resistant
- Termite-proof
- Sound and thermal insulation



Paver Blocks - Introduction

- ❖ Paver blocks are precast concrete or clay paving units used to create durable and aesthetically pleasing pavements.
- ❖ They are manufactured in different shapes, sizes, colors, and thicknesses to suit various applications.
- ❖ Paver blocks are laid over a prepared base without mortar, allowing easy installation and replacement.
- ❖ Key Characteristics
 - High compressive strength
 - Good drainage due to joints between blocks
 - Easy to install, remove, and replace
 - Resistant to weathering and abrasion
 - Available in various colors, shapes, and patterns





AAC(Autoclaved Aerated Concrete)Blocks – Introduction

- ❖ **Autoclaved Aerated Concrete (AAC) blocks** are lightweight, precast building blocks made from **cement, lime, sand (or fly ash), water, and aluminum powder**.
- ❖ The aluminum powder reacts with the other ingredients to form millions of tiny air pores, making the blocks lightweight and thermally efficient.
- ❖ The blocks are cured under **high-pressure steam in an autoclave**, which provides high strength and durability.
- ❖ Key Characteristics
 - Lightweight (about one-third the weight of conventional clay bricks)
 - Excellent fire resistance
 - Superior thermal insulation
 - Good sound insulation
 - High strength-to-weight ratio
 - Eco-friendly and energy-efficient





AAC (Autoclaved Aerated Concrete) Blocks – Advantages

- ❖ Reduces the dead load of structures
- ❖ Faster construction due to larger block size
- ❖ Reduces mortar and plaster consumption
- ❖ Excellent thermal and acoustic insulation
- ❖ High fire resistance (up to 4–6 hours depending on wall thickness)
- ❖ Resistant to pests, mold, and termites
- ❖ Environment-friendly with low construction waste



AAC (Autoclaved Aerated Concrete) Blocks – Disadvantages

- ❖ Higher initial cost than conventional clay bricks
- ❖ Brittle material; can crack if handled improperly
- ❖ Requires special AAC block adhesive instead of conventional mortar
- ❖ Lower load-bearing capacity compared to dense concrete blocks
- ❖ Requires proper anchors and fasteners for fixing heavy fixtures
- ❖ Water absorption is relatively high if not properly finished, requiring suitable plaster or waterproof coatings



CLC (Cellular Lightweight Concrete) Blocks – Introduction

- ❖ Cellular Lightweight Concrete (CLC) blocks are lightweight concrete blocks produced by mixing cement, sand, water, and pre-formed foam.
- ❖ The foam creates millions of tiny air voids within the concrete, significantly reducing its density and weight.
- ❖ Unlike AAC blocks, CLC blocks are not cured in an autoclave and are generally cured under normal conditions.
- ❖ Key Characteristics
 - Lightweight and easy to handle
 - Good thermal insulation
 - Good sound insulation
 - Excellent fire resistance
 - Environment-friendly and resource-efficient
 - Available in different densities based on application

Foam concrete



Aerated concrete





CLC (Cellular Lightweight Concrete) Blocks – Advantages

- ❖ Lower dead load on the structure
- ❖ Easy handling and faster construction
- ❖ Good thermal and acoustic insulation
- ❖ Excellent fire resistance
- ❖ Reduced transportation and labor costs
- ❖ Can utilize industrial by-products, making it eco-friendly
- ❖ Better workability than conventional concrete



CLC (Cellular Lightweight Concrete) Blocks – Disadvantages

- ❖ Lower compressive strength compared to AAC blocks and conventional concrete blocks
- ❖ Higher water absorption if not properly finished
- ❖ Shrinkage cracks may develop during drying
- ❖ Limited suitability for heavy load-bearing applications
- ❖ Requires careful quality control during foam generation and mixing
- ❖ Surface finishing may require additional plastering



Special Bricks Challenges

1. **Higher Initial Cost** – Special bricks such as AAC, CLC, refractory bricks, and decorative paver blocks are generally more expensive than conventional clay bricks.
2. **Transportation Issues** – Some special bricks require careful handling due to their lightweight or brittle nature, increasing transportation costs.
3. **Skilled Installation Required** – Proper installation often requires trained workers, special adhesives, or specific laying techniques.
4. **Limited Availability** – Availability may be restricted in rural or remote areas, leading to procurement delays.
5. **Application-Specific Use** – Each type of special brick is designed for specific purposes and may not be suitable for all construction applications.
6. **Moisture and Water Absorption** – Some lightweight bricks (e.g., AAC and CLC) require proper waterproofing and finishing to prevent moisture-related problems.
7. **Quality Control** – Performance depends on proper manufacturing, curing, and adherence to quality standards. Poor-quality products can lead to cracking and reduced durability.

THANK
YOU